



Secret Sector Z! Do this after the main mission. The challenges get harder as you go.

Stars

/5

1 Zeros at the end. Multiply the nonzero digits, then write the zeros from both factors.

a. $312 \times 8 =$ zeros: so $312 \times 80 =$

b. $423 \times 6 =$ zeros: so $4,230 \times 600 =$

2 Mental math or paper and pencil? Write M or P in each box.

a. 30×500

b. 468×37

c. $9 \times 6,000$

d. $2,053 \times 48$

e. Solve your M problems in your head. Write the letter, then the product.

Letter

=

Letter

=

f. Write a problem of your own that is good for mental math, and solve it:

.....

.....

3 Falco buys 48 energy cells at \$0.35 each. He says the total is \$168.00.

a. Find the real total.

Show your work

$$\begin{array}{r} \$0.35 \\ \times 48 \\ \hline \end{array}$$

Real total:

b. What mistake did Falco make?

.....

.....



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Name:

Date:

[illegible]

Test-level challenge! The Great Fox flies **6,284** miles each hour for **357** hours.

$$\boxed{} \times \boxed{} = \boxed{}$$

Show your work

6	2	8	4
×	3	5	7

YES

NO

The Great Fox makes **6** hyperspace jumps of **300,040** miles each. How far does it travel?

$$6 \times (\text{ } + \text{ }) = \text{ } + \text{ } = \text{ } \text{ miles}$$

◆ Bonus mission complete! You're ready for the test, pilot. ◆

**1. Multipliers ending in zeros: Lessons 2-3 and 2-9, pp. 70, 82**

a. $312 \times 8 = 2,496$, zeros: **1**, so $312 \times 80 = 24,960$

b. $423 \times 6 = 2,538$, zeros: **3**, so $4,230 \times 600 = 2,538,000$

In b, one zero comes from 4,230 and two from 600. Watch for 253,800, which counts only the zeros in 600.

2. Choose the method: p. 75; Chapter Test #30-31

a. **M** b. **P** c. **M** d. **P**

e. **a = 15,000** c = **54,000** (either order)

f. Accept any basic fact with zeros attached, solved correctly (for example, $70 \times 800 = 56,000$).

Rule of thumb: basic fact plus zeros is mental math; anything that needs regrouping or partial products is paper and pencil. For reference, b = 17,316 and d = 98,544.

3. Cents only: Lesson 2-10, p. 84

a. **$\$0.35 \times 48 = \16.80**

b. Falco put **the decimal point** in the wrong place: he counted one decimal place instead of two. *Accept any answer that names the decimal point.*

$35 \times 48 = 1,680$, then write the decimal point two places from the right. Estimate check: about $50 \times \$0.35 = \17.50 , so \$168 is far too much.

4. Test-level challenge: Lessons 2-5 and 2-8, pp. 74, 80

a. **$6,000 \times 400 = 2,400,000$** (6,284 rounds to 6,000; 357 rounds to 400)

b. **$6,284 \times 357 = 2,243,388$** miles c. **YES**

$6,284 \times 7 = 43,988$ $6,284 \times 50 = 314,200$ $6,284 \times 300 = 1,885,200$ total 2,243,388

Watch for 546,708: that comes from shifting the hundreds row ($6,284 \times 3$) only one place instead of two. If your child circles YES next to an answer like that, go over what "close" means.

5. Distributive Property: Lesson 2-2, p. 68; Chapter Test #29

a. **$6 \times (300,000 + 40) = 1,800,000 + 240 = 1,800,240$** miles

b. A full answer says: break 300,040 into $300,000 + 40$, multiply each part by 6, then add the two products. *Accept any explanation with those three steps.*